

CHAPTER – 4

ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION

The **Swiggy Instamart E-Commerce Database System** is designed to efficiently manage customers, products, categories, orders, payments, deliveries, and delivery partners. Functional dependencies help define the relationships between attributes in each table. Redundancy analysis and data anomaly detection ensure data consistency, reduce duplicate data, and improve database performance. A properly normalized database maintains integrity and supports efficient order processing.

1. CUSTOMER TABLE

Functional Dependency

Customer_ID → Customer_Name, Email, Phone_Number, Address

Redundancy Analysis

Customer details are stored only once in the Customer table. Orders store only the Customer_ID, avoiding duplicate customer information.

2. CATEGORY TABLE

Functional Dependency

Category_ID → Category_Name

Redundancy Analysis

Each product category is stored only once. Products reference Category_ID instead of repeating category names.

3. PRODUCT TABLE

Functional Dependency

Product_ID → Product_Name, Brand, Price, Stock, Category_ID

Redundancy Analysis

Product information is maintained only in the Product table. Orders refer to products using Product_ID, preventing duplicate product details.

4. ORDERS TABLE

Functional Dependency

Order_ID → Customer_ID, Order_Date, Total_Amount, Order_Status

Redundancy Analysis

The Orders table stores only order-related information. Customer details are accessed through Customer_ID.

5. ORDER_ITEM TABLE

Functional Dependency

Order_Item_ID → Order_ID, Product_ID, Quantity, Price

Redundancy Analysis

The Order_Item table links products with orders and avoids storing repeated product information.

6. PAYMENT TABLE

Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Status

Redundancy Analysis

Payment details are stored separately from orders, preventing duplicate payment information.

7. DELIVERY_PARTNER TABLE

Functional Dependency

Delivery_Partner_ID → Partner_Name, Phone_Number, Vehicle_Number

Redundancy Analysis

Delivery partner information is stored only once and linked to deliveries using Delivery_Partner_ID.

8. DELIVERY TABLE

Functional Dependency

Delivery_ID → Order_ID, Delivery_Partner_ID, Delivery_Status, Delivery_Time

Redundancy Analysis

Delivery details are stored separately from the Orders table, reducing duplicate delivery information.

Table	Data Anomalies	Reason
Customer	No significant anomalies	Customer details are stored only once in the Customer table. Orders reference the customer using Customer_ID , so updating or deleting an order does not affect customer data.
Category	No significant anomalies	Category information is stored separately. Products reference Category_ID , avoiding duplicate category names and preventing update anomalies.
Product	No significant anomalies	Product details are maintained only in the Product table. Changes to price or stock are made once, preventing duplicate or inconsistent product information.
Orders	No significant anomalies	The Orders table stores only order information. Customer and product details are accessed through foreign keys, eliminating redundant data.

Table	Data Anomalies	Reason
Order_Item	No significant anomalies	This junction table links Orders and Products. It avoids repeating product details in every order and maintains many-to-many relationships efficiently.
Payment	No significant anomalies	Payment details are stored in a separate table linked by Order_ID . Payment updates do not affect order or customer data, ensuring consistency.
Delivery_Partner	No significant anomalies	Delivery partner information is stored only once. Multiple deliveries can reference the same partner without duplicating partner details.
Delivery	No significant anomalies	Delivery details are maintained independently from Orders. Updating delivery status does not affect order, payment, or customer information.

CONCLUSION

The **Swiggy Instamart E-Commerce Database System** is designed using proper functional dependencies and normalized tables. Each table stores only relevant data, reducing redundancy and preventing **insert, update, and delete anomalies**. This database design improves **data consistency, integrity, efficiency, and overall system performance**, making it suitable for managing online grocery orders effectively.